

Vishal Yadav

Full-Stack Developer · Python | React | FastAPI | PostgreSQL

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PROFESSIONAL SUMMARY

Full-stack developer with a complete project portfolio spanning React frontends, FastAPI backends, and PostgreSQL databases. Experienced building real-time systems with WebSockets, designing normalized database schemas with Alembic migrations, and implementing JWT-based role access controls. Currently completing B.Tech (2026) and actively seeking a first full-time role where I can contribute across the entire development lifecycle.

TECHNICAL SKILLS

Languages	Python, JavaScript (ES6+), SQL
Frontend	React.js, Redux Toolkit, Axios, HTML5, CSS3, Tailwind CSS, DaisyUI
Backend	FastAPI, SQLAlchemy, REST API Design, WebSockets, JWT Authentication, Load Balancing basics
Database	PostgreSQL, Alembic Migrations, Database Schema Design, Query Optimisation
Tools	Git, GitHub, Postman, VS Code, Linux CLI
Concepts	System Design (fundamentals), MVC Architecture, Role-Based Access Control, API Rate Limiting

PROJECTS

Sync Music Platform · Personal Project²⁰²⁶

React · FastAPI · PostgreSQL · Redux · WebSockets · JWT Authentication | github.com/vishalyadav000232/sync-music-backend | [sync-music-frontend](#)

- Built a real-time collaborative music platform with WebSocket-powered synchronisation verified across up to 8 simultaneous connections, broadcasting host playback events (play, pause, seek, skip) with sub-second latency.
- Designed a room-based permission model with 2 roles (Host / Listener) and 5 exclusive host controls enforced at the WebSocket event level — not just the UI layer.
- Engineered scalable FastAPI WebSocket services to manage room membership, user presence, and event broadcasting; integrated REST APIs for auth and room CRUD with a clean separation of request/response and event-driven flows.
- Implemented JWT authentication with protected routes and token validation, enabling secure session management for all room participants.

Library Management System · Personal Project²⁰²⁵

React · FastAPI · PostgreSQL · SQLAlchemy · Alembic · JWT Authentication · Tailwind CSS | [library-backend](#) | [library-frontend](#)

- Designed and built 30+ RESTful API endpoints with FastAPI covering book management, user operations, seat allocation, and admin dashboard — organised into modular routers following separation-of-concerns principles.

- Modelled a 6-table normalised PostgreSQL schema with foreign key constraints and indexes; managed all schema evolution through Alembic versioned migrations, enabling reproducible zero-manual-SQL deployments.
- Implemented JWT-based authentication with 2 distinct roles (Admin / User), enforcing role-based access at the API dependency-injection layer and protecting all sensitive routes.
- Applied database optimisation techniques including strategic indexing and efficient relationship modelling to support scalable data retrieval across book, user, and seat allocation operations.

EDUCATION

Bachelor of Technology (B.Tech) · Dr. A.P.J Abdul Kalam Technical University *2022 – 2026 · Lucknow, India*

- Relevant coursework: Data Structures & Algorithms, Database Management Systems, Computer Networks, Operating Systems
- Final year project: Library Management System — full-stack application with React frontend, FastAPI backend, and PostgreSQL database

CERTIFICATIONS

- SQL and Relational Databases 101 — IBM (2025)
- Python Bootcamp Advanced — Udemy (2023)

All project source code available at github.com/vishalyadav000232